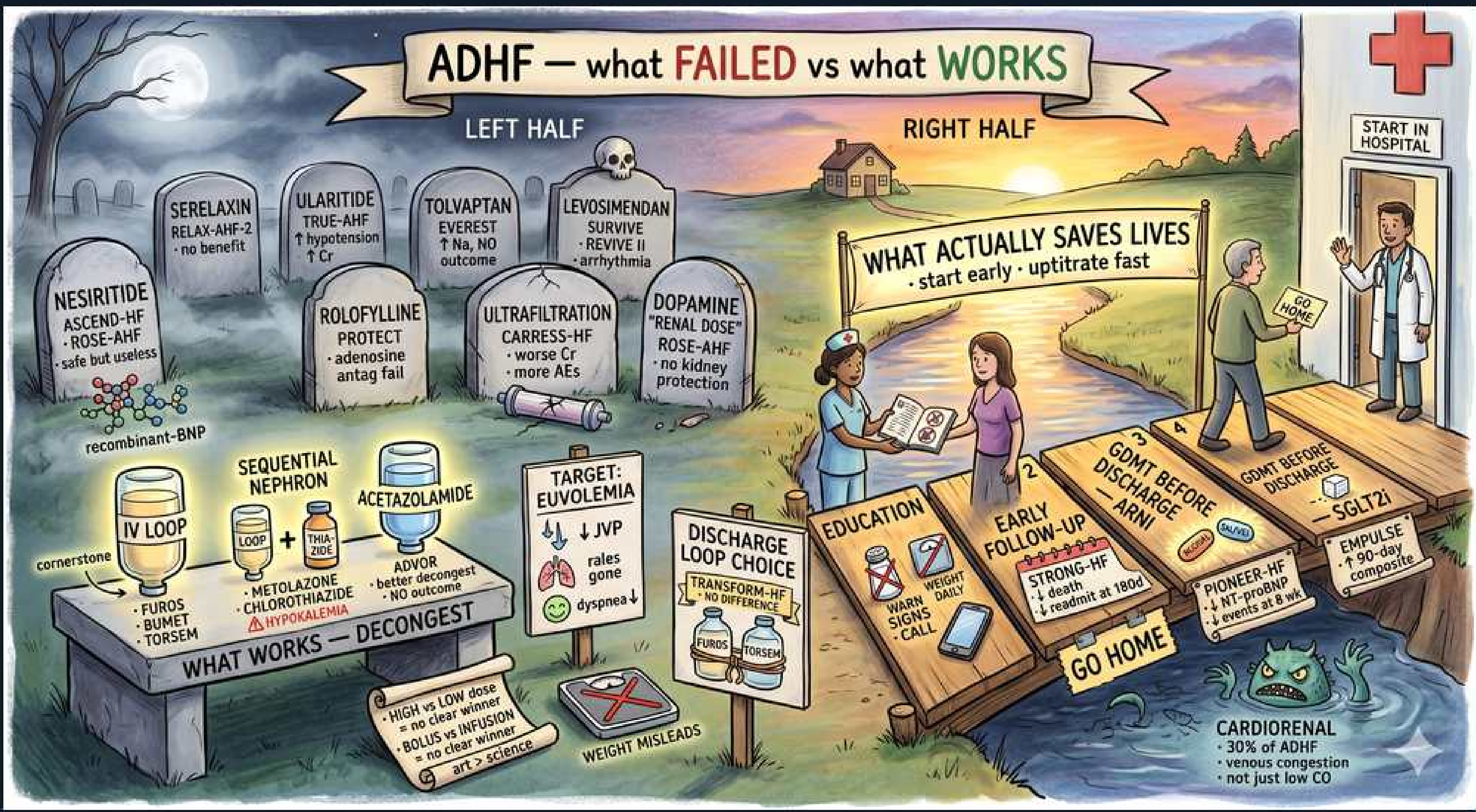
Heart Failure — SCENE 8: ADHF Trials & Discharge



1. Nesiritide — no benefit

WHAT YOU SEE

A tombstone in the LEFT-half graveyard reading 'NESIRITIDE — ASCEND-HF — safe but useless', with a recombinant-BNP molecule sketched beneath it — buried because it showed no outcome benefit.

WHAT IT ENCODES

Nesiritide is recombinant human BNP — a vasodilator that lowers PCWP and eases dyspnea modestly. The large ASCEND-HF trial (>7,000 patients) showed NO effect on death or 30-day rehospitalization and no clinically meaningful symptom benefit, with more hypotension. Earlier signals had even raised concern for worsening renal function. Bottom line: not part of routine ADHF care.

BOARD TRAP

WRONG answer: 'add nesiritide to improve survival and protect the kidneys in ADHF.' Discriminator: ASCEND-HF found no mortality/rehospitalization benefit and more hypotension — nesiritide is a dead-end drug, not a renoprotective or life-saving therapy.

---

2. Serelaxin — no benefit

WHAT YOU SEE

A graveyard tombstone reading 'SERELAXIN — RELAX-AHF-2 — no benefit' — recombinant relaxin-2 buried after its confirmatory trial was negative.

WHAT IT ENCODES

Serelaxin is recombinant human relaxin-2, a vasoactive peptide of pregnancy that early RELAX-AHF data suggested might cut mortality. The confirmatory RELAX-AHF-2 trial was NEGATIVE: no reduction in cardiovascular death or worsening heart failure. A reminder that promising phase-2 signals frequently die in phase 3.

BOARD TRAP

WRONG answer: 'serelaxin reduces cardiovascular mortality based on the RELAX-AHF program.' Discriminator: the early positive RELAX-AHF signal did NOT replicate — RELAX-AHF-2 was definitively negative, so serelaxin has no proven outcome benefit.

---

3. Ularitide — failed

WHAT YOU SEE

A graveyard tombstone reading 'ULARITIDE — TRUE-AHF — ↑hypotension, ↑Cr' — the urodilatin analog buried for improving surrogates but not outcomes.

WHAT IT ENCODES

Ularitide is a synthetic form of the natriuretic peptide urodilatin (vasodilator/natriuretic). The TRUE-AHF trial showed it reduced short-term hemodynamic strain and BNP but had NO effect on cardiovascular mortality or the long-term clinical course. Another natriuretic-peptide analog that improved surrogate markers without improving outcomes.

BOARD TRAP

WRONG answer: 'ularitide lowers BNP, therefore it improves survival in ADHF.' Discriminator: TRUE-AHF showed surrogate (BNP/hemodynamic) improvement does NOT translate to mortality benefit — a classic surrogate-vs-outcome trap.

---

4. Tolvaptan — EVEREST neutral

WHAT YOU SEE

A graveyard tombstone reading 'TOLVAPTAN — EVEREST — ↑Na, no outcome' — the V2 aquaretic buried because EVEREST was neutral for hard endpoints.

WHAT IT ENCODES

Tolvaptan is an oral V2 (vasopressin) receptor antagonist (aquaretic) that produces free-water clearance and RAISES serum sodium. The EVEREST trial showed it improves dyspnea, weight, and hyponatremia short-term but has NO effect on long-term mortality or HF rehospitalization. Useful niche: symptomatic hypervolemic hyponatremia; watch for overly rapid Na correction (osmotic demyelination).

BOARD TRAP

WRONG answer: 'start tolvaptan in ADHF to reduce mortality and prevent readmission.' Discriminator: EVEREST was neutral for hard outcomes — tolvaptan is reserved for symptomatic hyponatremia/congestion, not survival, and risks too-rapid sodium correction.

---

5. Levosimendan — no survival

WHAT YOU SEE

A graveyard tombstone reading 'LEVOSIMENDAN — SURVIVE / REVIVE II — ↑arrhythmia' — the calcium-sensitizer buried because it gave no survival benefit over dobutamine.

WHAT IT ENCODES

Levosimendan is a calcium-sensitizer inotrope plus K-ATP-channel vasodilator ('inodilator') that boosts contractility without raising myocardial O2 demand as much as catecholamines. The SURVIVE and REVIVE trials showed NO survival benefit vs dobutamine/placebo, with more hypotension and atrial fibrillation. Like all inotropes, it is for symptom/hemodynamic rescue, not life prolongation.

BOARD TRAP

WRONG answer: 'levosimendan improves survival over dobutamine because it is a calcium sensitizer.' Discriminator: SURVIVE showed no mortality difference vs dobutamine and more hypotension/AF — no inotrope (including levosimendan) prolongs life in ADHF.

---

6. Dopamine — no renal protection

WHAT YOU SEE

A graveyard tombstone reading 'DOPAMINE — "renal dose" — ROSE-AHF — no kidney protection' — the debunked low-dose-dopamine myth, buried by ROSE-AHF.

WHAT IT ENCODES

Low-dose ('renal-dose') dopamine ~2–3 mcg/kg/min was long believed to selectively dilate renal arteries and protect kidneys. The ROSE-AHF trial definitively BUSTED this myth: added to diuretics it gave NO improvement in decongestion (urine output) or renal function (cystatin C) — and dopamine caused more tachycardia. Same null result was seen with low-dose nesiritide in that trial.

BOARD TRAP

WRONG answer: 'add low-dose dopamine to protect the kidneys in a diuretic-refractory ADHF patient.' Discriminator: ROSE-AHF proved renal-dose dopamine does NOT enhance diuresis or preserve renal function — the 'renal dose' concept is a debunked myth.

---

7. IV loop diuretic — cornerstone

WHAT YOU SEE

The 'IV LOOP' diuretic bottle on the 'WHAT WORKS — DECONGEST' bench (left, alive side), labelled FUROS / BUMET / TORSEM and tagged 'cornerstone' — the undisputed decongestion mainstay.

WHAT IT ENCODES

IV loop diuretics (furosemide, bumetanide, torsemide) are the undisputed cornerstone of ADHF decongestion. The DOSE trial answered the dosing questions: continuous infusion was NOT superior to intermittent boluses, and a HIGH-dose strategy (~2.5× the home oral dose IV) relieved symptoms faster with greater diuresis and only transient, reversible bumps in creatinine vs low dose. Equivalences: furosemide 40 mg IV ≈ bumetanide 1 mg ≈ torsemide 20 mg; IV furosemide is ~2× as potent as oral.

BOARD TRAP

WRONG answer: 'continuous furosemide infusion is proven superior to bolus dosing and a small creatinine rise during diuresis means you must stop.' Discriminator: DOSE showed bolus = infusion and that high-dose diuresis works better despite a transient, benign creatinine rise — don't halt effective decongestion for that bump.

---

8. Sequential nephron blockade

WHAT YOU SEE

The 'SEQUENTIAL NEPHRON' add-on on the WHAT-WORKS bench: a LOOP bottle plus a THIAZIDE/metolazone-chlorothiazide bottle (with a red □ HYPOKALEMIA flag) — blocking a second nephron segment to beat diuretic resistance.

WHAT IT ENCODES

When a patient becomes diuretic-resistant (loops act only on the thick ascending limb, and chronic use causes distal-tubule hypertrophy with compensatory sodium reabsorption), add a thiazide-type agent for SEQUENTIAL nephron blockade — metolazone 2.5–5 mg or IV chlorothiazide blocks the distal convoluted tubule, restoring natriuresis. Give the thiazide ~30 min before the loop. Monitor closely for profound hypokalemia, hyponatremia, and volume depletion.

BOARD TRAP

WRONG answer: 'in diuretic resistance, just keep escalating the loop dose indefinitely.' Discriminator: distal-tubule sodium-avidity limits pure loop escalation — adding metolazone/thiazide (sequential blockade) overcomes resistance, but mandates aggressive electrolyte monitoring for hypokalemia/hyponatremia.

---

9. Acetazolamide (ADVOR)

WHAT YOU SEE

The 'ACETAZOLAMIDE' bottle on the WHAT-WORKS bench tagged 'ADVOR — better decongest' — the proximal-tubule add-on proven to enhance loop diuresis.

WHAT IT ENCODES

Acetazolamide is a carbonic-anhydrase inhibitor acting at the PROXIMAL tubule, blocking sodium-bicarbonate reabsorption. The ADVOR trial (2022) showed adding IV acetazolamide 500 mg/day to IV loop diuretics achieved successful decongestion in significantly more patients and shortened hospital stay. Bonus: it counteracts the contraction (metabolic) alkalosis caused by loops.

BOARD TRAP

WRONG answer: 'acetazolamide has no role in heart-failure decongestion and just causes metabolic acidosis.' Discriminator: ADVOR proved acetazolamide enhances loop-diuretic decongestion via proximal-tubule blockade and helps reverse diuretic-induced alkalosis — it is an evidence-based add-on, not merely an acidifier.

---

10. Target euvolemia

WHAT YOU SEE

The 'TARGET: EUVOLEMIA' sign on the works side showing '↓JVP, rales gone, ↓dyspnea' — full decongestion, not just symptom relief, is the endpoint.

WHAT IT ENCODES

The defined goal of ADHF therapy is full DECONGESTION to euvolemia, not just symptom relief — residual congestion at discharge is the strongest predictor of readmission and death. Track a falling JVP, resolving edema/crackles, declining daily weight, and improving dyspnea/orthopnea; a trending-down NT-proBNP supports adequate decongestion. Stopping diuretics once the patient 'feels better' but is still congested is a setup for early bounce-back.

BOARD TRAP

WRONG answer: 'discharge once dyspnea improves, even if the JVP is still elevated and there is residual edema.' Discriminator: residual congestion (persistently elevated JVP, weight not back to dry weight) is the #1 driver of 30-day readmission — euvolemia, not just symptomatic improvement, is the endpoint.

---

11. GDMT before discharge / ARNI

WHAT YOU SEE

A wooden plank on the 'BRIDGE HOME' reading 'GDMT BEFORE DISCHARGE — ARNI', with a PIONEER-HF tag — start guideline therapy in-hospital, don't defer.

WHAT IT ENCODES

Guideline-directed medical therapy (the 4 pillars: ARNI/ACEi/ARB, beta-blocker, MRA, SGLT2i) should be initiated and up-titrated DURING the hospitalization, once the patient is hemodynamically stable and off IV vasoactives. PIONEER-HF showed in-hospital initiation of sacubitril/valsartan (ARNI) is safe and lowers NT-proBNP more than enalapril, with fewer rehospitalizations. The hospital stay is a key 'teachable moment' window — don't defer.

BOARD TRAP

WRONG answer: 'wait weeks after discharge and full recovery before starting any GDMT.' Discriminator: PIONEER-HF and STRONG-HF support starting/intensifying GDMT before discharge — delaying it forfeits the highest-risk early window and increases readmission/death.

---

12. SGLT2i in-hospital (EMPULSE)

WHAT YOU SEE

An 'SGLT2i' pill on a 'BRIDGE HOME' plank tagged 'EMPULSE — 90-day composite' — start the SGLT2 inhibitor before discharge regardless of EF or diabetes.

WHAT IT ENCODES

SGLT2 inhibitors (empagliflozin, dapagliflozin) are a pillar of GDMT that work regardless of EF or diabetes status. EMPULSE showed starting empagliflozin in hospitalized ADHF patients improved a composite of death, HF events, and quality of life within 90 days; SOLOIST-WHF (sotagliflozin started at/just after discharge) reduced CV death and HF hospitalizations. They add early decongestion (osmotic/natriuretic diuresis) plus durable outcome benefit.

BOARD TRAP

WRONG answer: 'SGLT2 inhibitors only help diabetic heart-failure patients and must wait until the outpatient setting.' Discriminator: EMPULSE/SOLOIST/DAPA-HF/EMPEROR show benefit across EF AND independent of diabetes, with safe in-hospital initiation — withholding them for non-diabetics or until discharge is wrong.

---

13. Early follow-up + education

WHAT YOU SEE

The 'EDUCATION' (nurse with a teaching book: weigh daily, warn signs, calls) and 'EARLY FOLLOW-UP' planks on the bridge home, tagged STRONG-HF — the transition-of-care bundle that cuts readmission.

WHAT IT ENCODES

Transition-of-care bundles cut readmissions: patient education (daily weights, salt/fluid limits, medication adherence, symptom action plan) plus EARLY follow-up within 7–14 days. The STRONG-HF trial showed an intensive strategy of rapid GDMT up-titration with frequent early visits (labs/exam at ~1, 2, 3, 6 weeks) significantly reduced 180-day death or HF readmission versus usual care.

BOARD TRAP

WRONG answer: 'routine 4–6 week follow-up with slow, cautious up-titration is the safest post-ADHF strategy.' Discriminator: STRONG-HF proved a high-intensity early follow-up with rapid up-titration is SAFER and reduces death/readmission — delayed, leisurely titration is the inferior approach.

---

14. Cardiorenal — venous congestion

WHAT YOU SEE

The angry green 'CARDIORENAL' kidney creature in the water under the bridge, tagged '30% of ADHF — venous congestion — not just low CO' — congestion (not low output) drives most cardiorenal syndrome.

WHAT IT ENCODES

Worsening renal function complicates ~25–30% of ADHF admissions. The key teaching point: in congested patients the dominant driver of cardiorenal syndrome is elevated central VENOUS pressure (renal venous congestion raising intrarenal/interstitial pressure and dropping the trans-renal perfusion gradient), more than low forward cardiac output. Therefore effective DECONGESTION often IMPROVES renal function, and a small creatinine rise during successful diuresis ('pseudo-worsening') is usually benign.

BOARD TRAP

WRONG answer: 'a rising creatinine during diuresis means the kidneys are being underperfused — stop the diuretic and give fluids.' Discriminator: venous congestion, not low output, is the main cause of cardiorenal syndrome in congested ADHF — continuing decongestion usually improves renal function, whereas fluids/stopping diuretics worsen congestion.
